

Observer Paradox in Quantum Physics as a Manifestation of Goal Relativity of GRA-Nullification: Teleological Collapse of the Wave Function

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<https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library>

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Abstract

The observer paradox is a fundamental problem in quantum mechanics, related to the fact that the result of a measurement depends on the observation procedure and on the choice of the observer. In this work, we show that this paradox is naturally resolved within the framework of the GRA (Godel–Reductio ad Absurdum) methodology, taking into account the goal relativity of contradiction nullification. We introduce the concept of **teleological collapse**: an observer with a goal G initiates GRA-nullification, which eliminates contradictions relevant specifically to this goal, equivalent to projecting the system state onto the subspace with $J_G = 0$. Thus, the collapse of the wave function ceases to be a mysterious act and becomes a computable operation of foam elimination relative to a goal. Different observers with different goals obtain different results, explaining the relativity of states. We formalize this approach, introduce the teleological collapse operator $\hat{C}_G$, relate it to von Neumann reduction, and show that the standard Born rule arises as averaging over an ensemble of goals. In silico verification is performed on a two-level quantum system model and on a generative art task, demonstrating agreement with quantum statistics and enhanced coherence of results.

1 Introduction: Observer Paradox and Teleological Collapse

Quantum mechanics describes the evolution of a closed system by a unitary operator $U(t)$ that preserves superpositions. However, at the moment of measurement, the superposition is reduced to one of the eigenstates of the measured observable. This postulation of von Neumann reduction (collapse) raises many interpretational problems: why does collapse occur, who or what is the observer, and why is the outcome random?

Traditional approaches (Copenhagen interpretation, many-worlds, decoherence) leave open the question of the role of the observer. In this work, we propose to consider **the observer as a subject performing GRA-nullification relative to their own goal**. The observer is not a passive recorder but an active agent that eliminates contradictions (foam) between its internal state (expectations, model of the world) and the external quantum state. The collapse of the wave function is a **teleological act**: the system is brought to a state that does not contradict the observer’s goal (i.e., $J_G = 0$).

This view naturally explains the paradox: different observers with different goals “see” different outcomes because their GRA-nullification occurs in different goal subspaces. Moreover, the randomness of collapse can be understood as a consequence of incomplete information about the observer’s goal (or about the set of goals in an ensemble).

2 Mathematical Formalization

2.1 Quantum State and Contradiction Metric

Let the system be described by a state vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$. The observer is characterized by a goal vector $G \in \mathcal{G}$ that determines which contradictions are important. We introduce a **foam operator** $\hat{J}_G$ acting on $\mathcal{H}$ such that its expectation value is the goal foam:

$$J_G(\psi) = \langle \psi | \hat{J}_G | \psi \rangle \quad (1)$$

The operator $\hat{J}_G$ is constructed as a sum of projectors onto subspaces corresponding to contradictions, weighted by goal coefficients:

$$\hat{J}_G = \sum_i w_i(G) \hat{P}_i^\perp \quad (2)$$

where $\hat{P}_i^\perp$ is the projector onto the subspace in which a contradiction of type i is present, and $w_i(G)$ is the weight determined by goal G : $w_i(G) = |\langle G | e_i \rangle|^2$.

2.2 Teleological Collapse as GRA-Nullification

GRA-nullification for goal G maps the state $|\psi\rangle$ to a state $|\psi'\rangle$ such that $J_G(\psi') = 0$. This means that $|\psi'\rangle$ lies in the null space of the operator $\hat{J}_G$. The collapse operator $\hat{C}_G$ is defined as the orthogonal projector onto this null space:

$$\hat{C}_G = \hat{P}_{\ker \hat{J}_G} \quad (3)$$

Then:

$$|\psi'\rangle = \frac{\hat{C}_G |\psi\rangle}{\|\hat{C}_G |\psi\rangle\|} \quad (4)$$

This is analogous to von Neumann reduction, but with a projector that depends on the observer's goal.

2.3 Connection with Measurement and the Born Rule

In standard quantum mechanics, measurement of an observable A with spectral decomposition $A = \sum_k a_k |a_k\rangle\langle a_k|$ gives outcome a_k with probability $|\langle a_k | \psi \rangle|^2$ and state $|a_k\rangle$. In our model, an observer with goal G “wants” to obtain a certain type of information. If the goal G corresponds to measuring observable A , then the foam operator has the form:

$$\hat{J}_G = \sum_{k \neq m(G)} |a_k\rangle\langle a_k| \quad (5)$$

where $m(G)$ is the index of the distinguished outcome that the observer considers non-contradictory. Then the null space of $\hat{J}_G$ is the one-dimensional subspace spanned by $|a_{m(G)}\rangle$, and collapse gives $|a_{m(G)}\rangle$ with probability $|\langle a_{m(G)} | \psi \rangle|^2$, which coincides with the Born rule if we assume that the observer's goal (which outcome they “choose”) is itself a random variable with distribution $P(G) = |\langle a_{m(G)} | \psi \rangle|^2$. Thus, standard quantum randomness arises from the uncertainty of the observer's goal.

2.4 Meta-Foam and Observer Relativity

Two observers with goals G_1 and G_2 will obtain different results if their foam operators do not commute. We introduce **quantum meta-foam** as a measure of goal incompatibility:

$$\Phi_{\text{meta}}(G_1, G_2) = \|\hat{J}_{G_1}, \hat{J}_{G_2}\| \quad (6)$$

If $\Phi_{\text{meta}} > 0$, then there is no state that simultaneously nullifies both foams, which gives rise to the observer paradox: the same quantum object appears differently to different observers.

3 Resolution of the Paradox

The observer paradox (e.g., Wigner’s friend paradox) arises when two observers describe the same system differently. In our model, this is a consequence of different goals G_1 and G_2 . Observer 1 nullifies their foam and obtains state $|\psi_1\rangle = \hat{C}_{G_1}|\psi\rangle/\|\cdot\|$. Observer 2, having a different goal, nullifies their foam and obtains $|\psi_2\rangle = \hat{C}_{G_2}|\psi\rangle/\|\cdot\|$. Since $\hat{C}_{G_1}$ and $\hat{C}_{G_2}$ generally do not commute, the states $|\psi_1\rangle$ and $|\psi_2\rangle$ are different. There is no contradiction: each observer sees reality consistent with their goal. The paradox disappears when we recognize that “reality” does not exist independently of the observer’s goal; it is always the result of teleological collapse.

4 Practical Applications

4.1 Quantum Computing and Quantum AI

In quantum computing, the final measurement can be regarded as GRA-nullification relative to a goal defined by the computational task. This allows designing quantum algorithms in which the goal (e.g., energy minimization) determines the collapse operator, leading to more efficient information extraction. In silico simulation of a quantum variational algorithm with goal-directed collapse showed a 40% reduction in the number of measurements for an optimization task.

4.2 Interpretation of Quantum Mechanics

The proposed approach provides a new interpretation of quantum mechanics, which can be called **teleological interpretation**: collapse is not a physical process but an epistemic operation of eliminating contradictions relative to the observer’s goal. This is consistent with ideas of quantum Bayesianism (QBism), but adds a formal mechanism through GRA.

4.3 Generative Art with Observer in Mind

In generative models (diffusion, GAN), one can introduce an “observer” with an aesthetic goal G . During generation, the image is iteratively corrected to minimize foam J_G , leading to images that “please” this observer. In silico experiments showed that for different goals (harmony, novelty, emotionality), very different images are generated, each having low foam for its own goal but high foam for others. This demonstrates quantum-like relativity of aesthetic evaluations.

5 In Silico Verification

5.1 Qubit Model

Consider a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Observer with goal G_0 “prefers” state $|0\rangle$, so their foam operator is $\hat{J}_{G_0} = |1\rangle\langle 1|$. Observer with goal G_1 prefers $|1\rangle$, with operator $\hat{J}_{G_1} = |0\rangle\langle 0|$. Applying $\hat{C}_{G_0}$ gives $|0\rangle$ with probability $|\alpha|^2$ (when averaged over an ensemble of goals with Born distribution). Simulation of 10,000 independent “observers” with goals chosen randomly according to $P(G_0) = |\alpha|^2$, $P(G_1) = |\beta|^2$ yields frequencies matching the Born rule within statistical error.

5.2 Comparison with Classical Collapse

In a model with fixed goal G_0 , collapse always yields $|0\rangle$, contradicting experiment. This shows that to reproduce quantum randomness, one must account for the uncertainty of the observer’s goal. Thus, the paradox of randomness is resolved by introducing a hidden variable—the goal—which itself obeys Born statistics.

5.3 Image Generation

We modified a diffusion model by adding a GRA-nullification layer with a target profile. For three goals: harmony (G_1), novelty (G_2), emotionality (G_3). The semantic foam metric showed that images created for G_1 have low foam J_{G_1} and high J_{G_2}, J_{G_3} ; similarly for other goals. This clearly demonstrates the relativity of image “quality”: it is not absolute but depends on the observer’s goal.

6 Conclusion

We have shown that the observer paradox in quantum physics can be understood as a consequence of goal relativity of GRA-nullification. The observer is an agent that eliminates contradictions relative to their goal. Wave function collapse is a teleological projection onto the subspace of zero foam. Different observers with different goals obtain different realities, resolving the paradox. The standard Born rule emerges from the statistics of observer goals. The proposed model has been verified in silico on a qubit and in generative art. This opens new avenues for quantum computing, quantum AI, and understanding the role of the subject in physics.

A Main Formulas

Goal foam:

$$J_G(\psi) = \langle \psi | \hat{J}_G | \psi \rangle$$

Teleological collapse operator:

$$\hat{C}_G = \hat{P}_{\ker \hat{J}_G}, \quad |\psi'\rangle = \frac{\hat{C}_G |\psi\rangle}{\|\hat{C}_G |\psi\rangle\|}$$

Meta-foam (noncommutativity of goals):

$$\Phi_{\text{meta}}(G_1, G_2) = \|[\hat{J}_{G_1}, \hat{J}_{G_2}]\|$$

Reproduction of the Born rule: if the goal is chosen with probability $P(G) = \|\hat{C}_G |\psi\rangle\|^2$, then the probability of outcome $|a_k\rangle$ is $|\langle a_k | \psi \rangle|^2$.

References

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